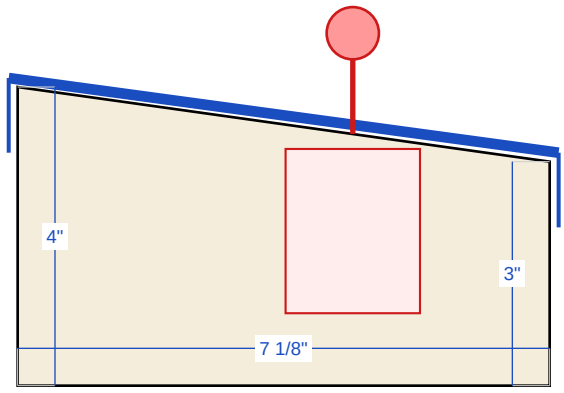
SUPERCADE control box

Try the controls first: a tabletop box you plug into a laptop or TV. The deck plate moves into the cabinet later.

SIDE SECTION (players on the right)



tan = tray (B1) blue = lid (B2) + Foamex plate (P12) red = joystick

PARTS

B1 **Tray**
22 1/4" x 14 1/8" cardboard, one piece (page 2)

B2 **Lid**
16 1/2" x 9 3/8" cardboard, one piece (page 3)

P12 **Deck plate**
14 1/2" x 7 3/8" 5 mm Foamex - the same plate the cabinet uses

ALSO NEED

- The 2-player zero-delay kit (2 sticks, 16 x 24 mm buttons, 2 encoders)
- 4 x M4/M5 bolts + nuts for the plate corners, 8 for the sticks
- ~10 kit screws + nuts, glue, tape
- 1 kg of weight inside (bag of rice) + a non-slip mat underneath, so the box doesn't walk when someone yanks a stick
- A laptop, or a TV + Raspberry Pi

WHY THIS SHAPE

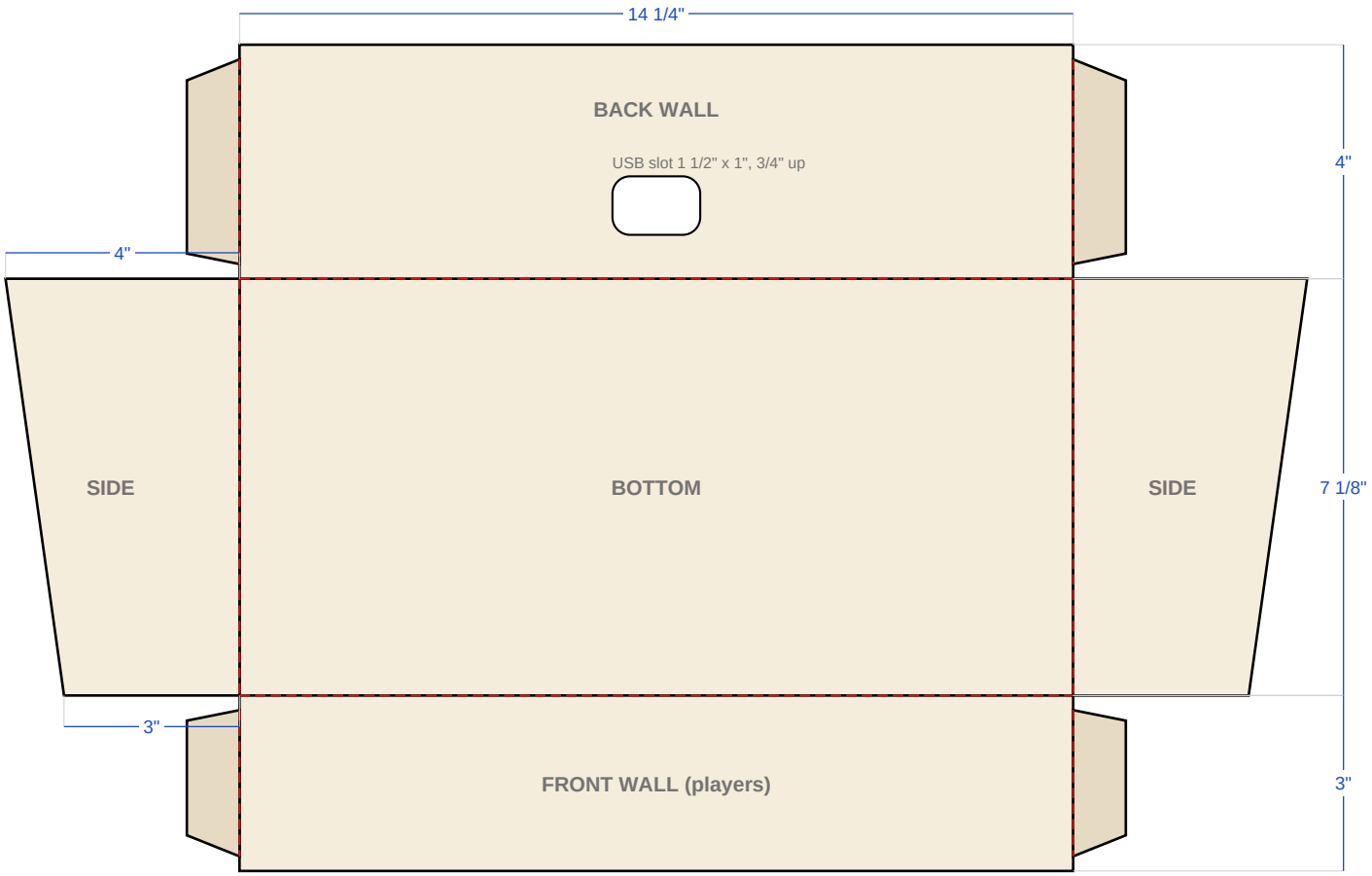
It's a shoebox with a sloped top. The lid is the control deck, and its skirts fold DOWN over the outside of the tray, so nothing folds inward into the space the joysticks and buttons need. The slope (back 4", front 3") gives the same ~7 degree tilt as the cabinet's deck. The Foamex plate is exactly the cabinet's P12, and the template is the same one: if the kids love it, unbolt the plate with everything still wired and drop it into the cabinet.

CHECK FIRST

Measure your joystick from the underside of its mounting plate to the bottom of its microswitches / connector. The box leaves about 2 3/4" at the stick. If yours needs more, add the same amount to BOTH wall heights (and to the side walls) before cutting.

B1 Tray - one piece

Cut on the black lines, score + fold UP on the red dashed lines. Measurements in inches.



Blank: 22 1/4" x 14 1/8". Draw the 14 1/4" x 7 1/8" BOTTOM first, then add the walls around it.

SIDE walls are sloped: 4" tall at the back edge, 3" at the front. Draw them as a line out square from each corner of the bottom, then join the ends.

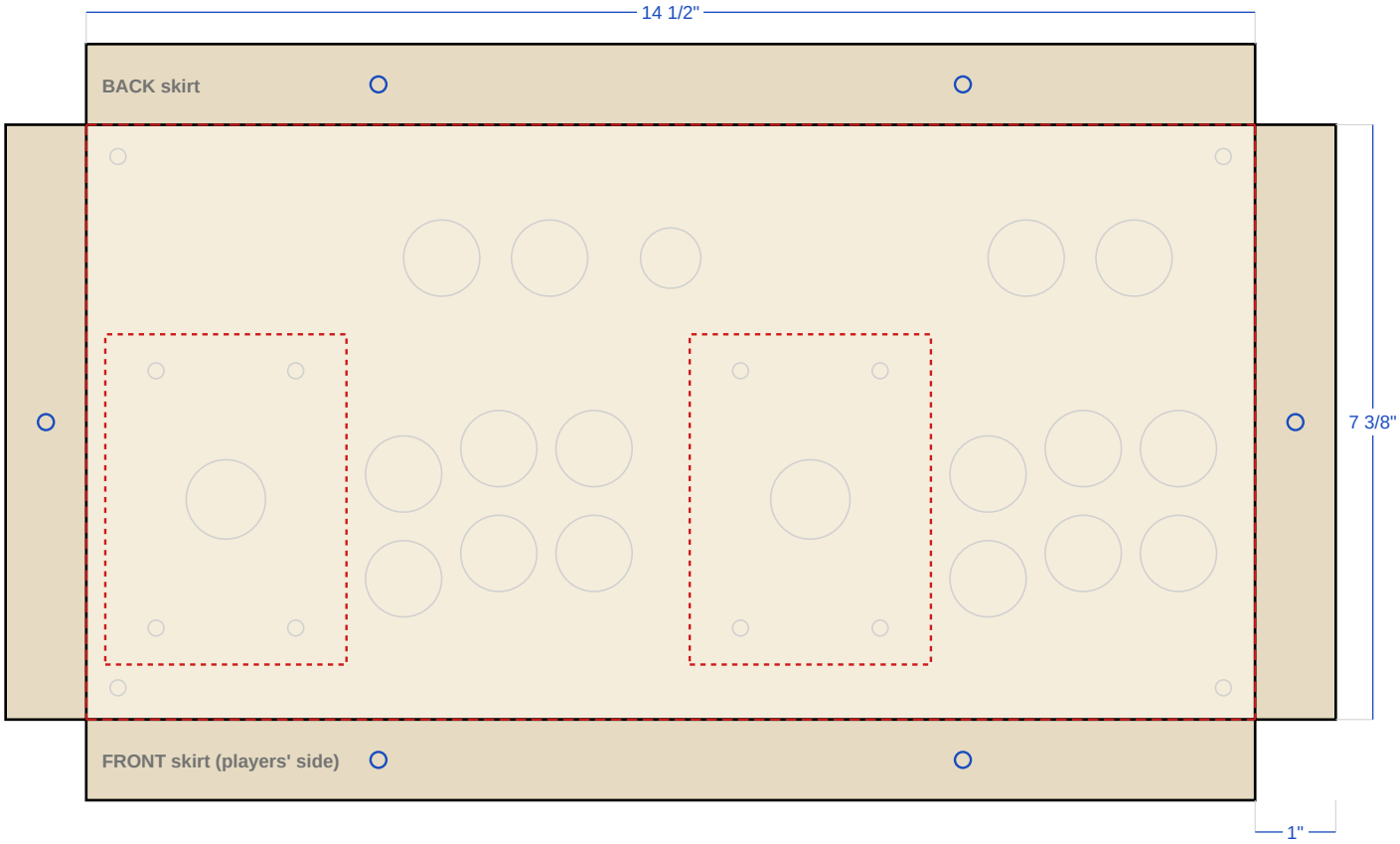
Corner TABS (7/8" wide, shaded) are on the front and back walls. Trim their outer corners at a slant so they clear the folds. They fold round INSIDE the side walls: glue them, then put one kit screw + nut through each tab and side wall.

The USB slot takes both encoder cables out of the back. Cut it before folding.

Fold order: back + front walls up, tabs in, side walls up over the tabs.

B2 Lid = control deck

Cut on black, score + fold DOWN on red. The Foamex plate (P12) bolts on top, lined up with the edges.



Blank: 16 1/2" x 9 3/8". The top is exactly the deck plate (14 1/2" x 7 3/8"); the 1" skirts fold down and the four corner squares are cut away.

Holes: tape the 1:1 control-panel template (templates/control-panel-1to1.pdf) onto the FOAMEX plate and cut every hole there. Then lay the plate on the lid, mark through, and cut the lid's button and bolt holes.

Red dashed boxes: cut these right out of the lid, under each joystick. They're sized for a typical stick plate (about 3" x 4"); make them just bigger than YOUR plate. The stick bolts to the Foamex alone, which keeps the panel thin enough for the shaft.

Blue circles: one kit screw + nut through skirt and tray wall at each (6 total). Keep the lid unglued so you can open it to fix wiring.

Bolt the plate down through the 4 corner holes (plate + lid only).

Build + wire + play

About an afternoon. Kids can do the folding, screwing and button-pressing; adults do the knife work.

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1. **Cut the plate first**
   Tape the 1:1 template to the 5 mm Foamex, prick every crosshair, cut the holes (hole saw / step bit, or compass + knife). This is the part that has to be neat.

2. **Cut + fold the tray (B1)**
   Page 2. Cut the USB slot, fold the walls up, glue + screw the corner tabs.

3. **Cut the lid (B2)**
   Page 3. Mark the holes through the plate, cut them and the two joystick openings, fold the skirts.

4. **Mount the controls on the bench**
   Bolt the plate onto the lid. Screw the buttons in from the top (nut underneath), bolt each stick on from below.

5. **Wire it**
   Each encoder board: joystick harness into the stick, one 2-wire lead per button. Player 1 on one board, Player 2 on the other. Stick the boards to the tray bottom with double-sided tape, and run both USB cables out of the slot.

6. **Close it up**
   Put the 1 kg weight in, drop the lid on, 6 screws through the skirts. Non-slip mat underneath.

7. **Test**
   Plug into a laptop: on Windows, 'Set up USB game controllers'; on Mac/Linux, any online gamepad tester. Every button should light up once. Swap any lead that shows up wrong.

8. **Play**
   Easiest start: a Raspberry Pi with an arcade front-end (e.g. RetroPie / Batocera) on the TV. It sees the box as two gamepads. Pick a handful of 2-player games you're happy with.

IS IT WORTH THE CABINET? A FEW WEEKS OF...

- Do they ask for it, or only play when it's offered?
- Do they play together (good sign) or fight over it (fixable: take turns by credits)?
- Does it fit into the day without pushing out sleep, outside play or homework?

Credits trick: the COIN buttons are a built-in limit - e.g. '5 credits after homework'.
If it's a hit: unbolt the plate with everything still wired and move it to the cabinet's control deck.